Sketch2Face

Cross-modal face retrieval

CUFS CUHK subset

Ipsita | Roll: 12 308 012 (placeholder)

B.Tech in Artificial Intelligence and Machine Learning

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Guide: [Guide name - placeholder]

Problem and objective

Input: one face sketch and a gallery of candidate photos

Output: photos ranked by similarity to the person in the sketch

Challenge: preserve identity across pencil strokes and photographic appearance

Objective: build the matcher and measure rank-k retrieval accuracy

Dataset audit

Full CUFS description: 606 identities across CUHK, AR, and XM2VTS

Downloaded mirror: 188 usable CUHK sketch-photo pairs

Extra unpaired sketches and duplicate original files were excluded

Every reported result is for the verified CUHK subset

Identity-disjoint evaluation

114 training identities: learn sketch-photo correspondence

37 validation identities: select the checkpoint

37 test identities: final held-out experiment

Each test sketch searches all 37 held-out photos

Model architecture

Stage	Operation	Output
Preprocessing	Grayscale, resize, normalize	3 x 224 x 224
Modality input	Sketch or photo residual adapter	3 x 224 x 224
Shared encoder	ImageNet-pretrained ResNet-18	512 features
Projection	Linear, ReLU, Linear, normalize	256-D embedding
Retrieval	Cosine similarity to gallery	Ranked photos

Paired contrastive learning

A batch contains sketches and photos of distinct paired identities

Correct sketch-photo pairs are positives; other people are negatives

Learn in both directions: sketch to photo and photo to sketch

$L = 0.5 \times [CE(S\ P^T / \tau) + CE(P\ S^T / \tau)],\ \tau = 0.07$

Training and comparisons

20 epochs; batch size 32; AdamW optimization

Freeze backbone for 3 epochs, then fine-tune the final residual stage

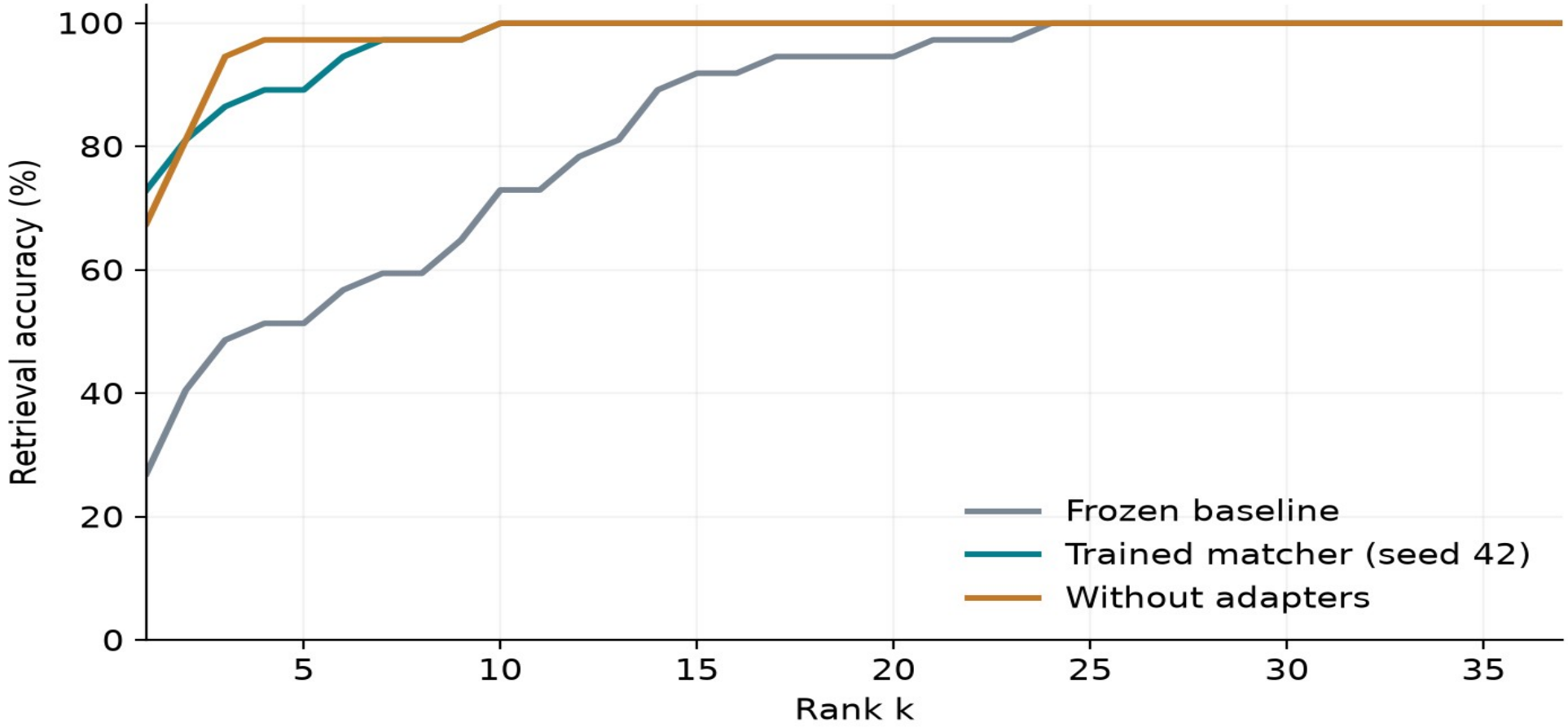
Compare frozen baseline, trained matcher, and no-adapter ablation

Repeat matcher training with seeds 42, 43, and 44

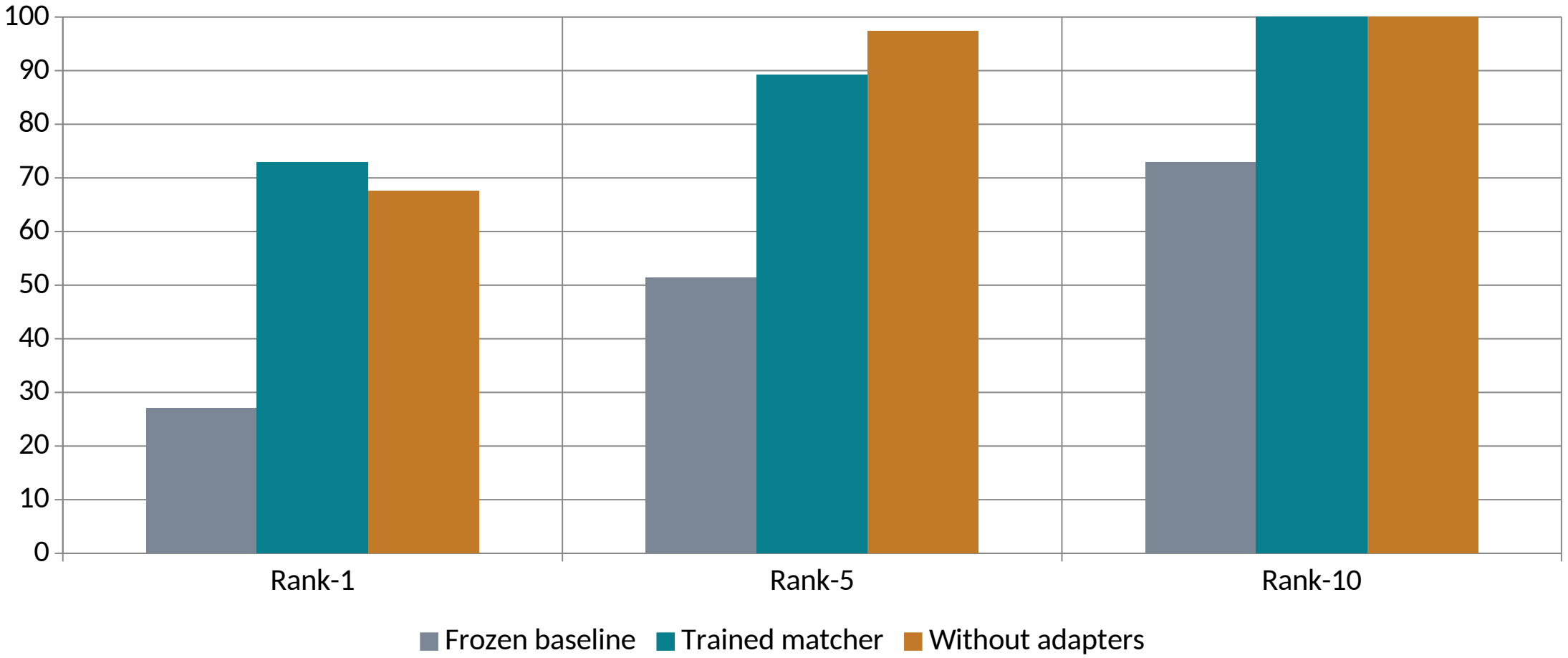
Measured retrieval accuracy

Model	Epoch	R1 (%)	R5 (%)	R10 (%)
Frozen baseline	0	27.03	51.35	72.97
Matcher seed 42	20	72.97	89.19	100.00
Matcher seed 43	20	67.57	89.19	100.00
Matcher seed 44	16	67.57	91.89	100.00
Without adapters	18	67.57	97.30	100.00

CMC comparison



Rank-k comparison



What the experiments show

Primary rank-1: 27 correct queries out of 37 (72.97%)

Frozen baseline: 10 correct queries out of 37 (27.03%)

Paired rank-1 gain: 45.95 percentage points; 95% interval 27.03–64.86

Three-seed mean rank-1: 69.37% ± 3.12 percentage points

Adapter ablation and errors

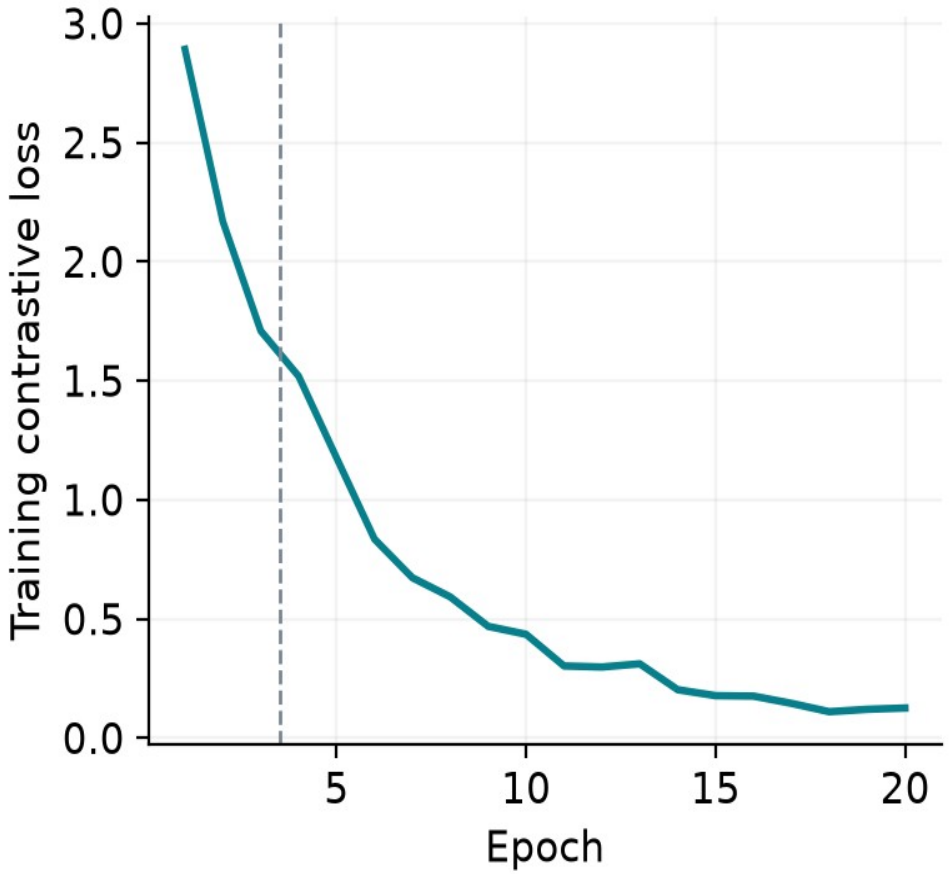
Without adapters: 67.57% rank-1, 97.30% rank-5

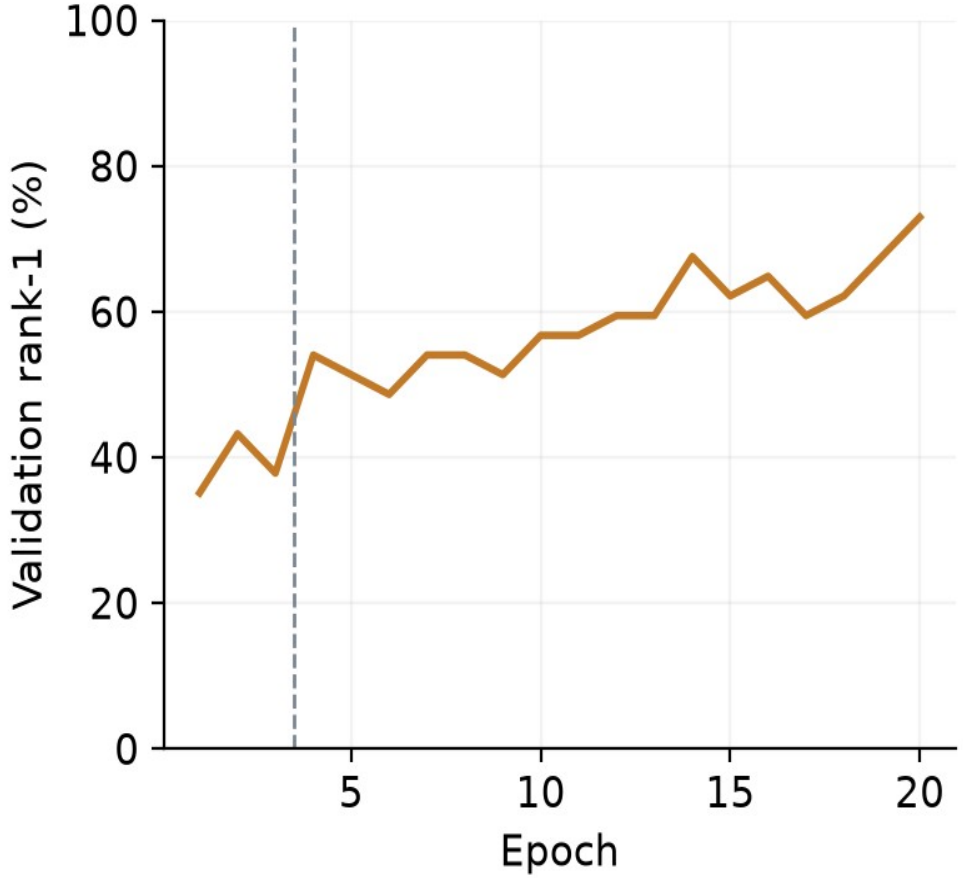
Adapters improve primary rank-1, but reduce rank-5 in this comparison

Ten primary queries fail at rank 1; the hardest correct match is rank 10

One ablation seed does not establish a consistent adapter benefit

Learning curves





Limitations

Only the CUHK subset; AR and XM2VTS pairs remain untested

Small closed-set gallery: 37 photos; no unknown-person rejection

Viewed artist sketches differ from eyewitness and composite sketches

No demographic fairness, forensic reliability, or state-of-the-art claim

Demo and conclusion

Run: uv run streamlit run app.py

Select a held-out sketch, rank the gallery, and inspect the correct rank

Paired training substantially improves the measured frozen baseline

Next: full CUFS, larger galleries, repeated ablations, forensic datasets

References

Wang & Tang, Photo-Sketch Synthesis and Recognition, TPAMI 2009

Zhang et al., Coupled Information-Theoretic Encoding, CVPR 2011

He et al., Deep Residual Learning, CVPR 2016

Radford et al., Contrastive multimodal learning, ICML 2021

Loshchilov & Hutter, AdamW, ICLR 2019; CUFS mirror, Kaggle v1